

TORBAT-HEYDARIEH, Iran, Islamic Republic of

WMO: 407620

Lat: 35.3315N

Lon: 59.2059E

Elev: 4760

StdP: 12.34

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 14.6	(c) 19.6	(d) -1.3	(e) 6.1	(f) 28.3	(g) 4.2	(h) 8.1	(i) 31.9	(j) 20.6	(k) 35.9	(l) 17.8	(m) 34.4	(n) 3.9	(o) 330	(p) 0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.3	(c) 95.2	(d) 61.9	(e) 93.0	(f) 61.2	(g) 91.0	(h) 60.5	(i) 65.7	(j) 89.3	(k) 64.1	(l) 88.1	(m) 62.7	(n) 86.9	(o) 11.0	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.3	80.5	80.2	53.9	73.7	77.8	51.8	68.3	75.7	33.3	89.5	31.9	88.0	30.7	87.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.2	(b) 18.8	(c) 16.7	(e) 8.8	(f) 99.9	(g) 5.4	(h) 2.0	(i) 4.9	(j) 101.4	(k) 1.8	(l) 102.6	(m) -1.3	(n) 103.7	(o) -5.2	(p) 105.1		
(a) 22.2	(b) 18.8	(c) 16.7	(e) 7.9	(f) 66.8	(g) 5.2	(h) 2.8	(i) 4.1	(j) 68.9	(k) 1.1	(l) 70.5	(m) -1.9	(n) 72.1	(o) -5.7	(p) 74.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.4	34.4	38.4	47.7	58.3	69.1	77.5	80.9	78.2	70.4	59.6	46.7	38.2
	DBStd	17.43	7.32	7.46	7.81	7.81	6.28	4.86	3.98	4.03	5.54	6.81	6.98	5.98
	HDD50	1487	483	329	137	21	0	0	0	0	0	7	141	369
	HDD65	4077	948	746	539	221	31	1	0	0	17	193	549	832
	CDD50	4549	0	4	64	271	591	824	958	875	613	304	43	2
	CDD65	1662	0	0	1	21	157	375	493	410	180	25	0	0
	CDH74	18184	0	0	16	216	1512	4138	5719	4354	1902	325	2	0
	CDH80	7710	0	0	1	38	490	1838	2746	1886	655	56	0	0
Wind	WSAvg	6.4	3.6	4.8	5.8	6.7	7.7	8.8	10.2	9.1	6.4	5.4	4.3	3.6
Precipitation	PrecAvg	10.4	1.5	1.7	2.3	1.6	1.0	0.2	0.0	0.0	0.1	0.3	0.7	1.1
	PrecMax	16.7	3.2	5.2	5.7	3.5	3.4	1.6	0.9	0.1	0.7	2.0	1.7	3.7
	PrecMin	4.0	0.3	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.3	0.8	1.3	1.5	1.0	0.8	0.4	0.2	0.0	0.2	0.5	0.5	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.4	63.5	76.1	84.7	91.8	97.2	98.8	96.1	92.8	85.2	72.4	62.0
		MCWB	42.9	45.7	52.1	55.3	59.0	62.1	63.1	62.3	59.0	55.9	51.3	44.4
	2%	DB	53.5	57.7	69.8	79.5	88.2	94.1	96.0	93.2	89.4	80.9	67.4	56.4
		MCWB	41.2	43.7	49.9	55.0	58.4	60.9	62.5	61.2	58.0	54.6	48.5	42.0
	5%	DB	49.7	54.1	65.9	76.0	85.4	91.6	93.6	90.9	86.5	77.6	63.8	53.0
		MCWB	39.4	42.1	48.2	54.3	57.8	60.6	62.0	60.5	57.4	53.4	46.8	40.5
	10%	DB	46.2	50.9	61.9	72.5	82.2	89.2	91.4	88.8	83.9	74.0	60.2	49.5
		MCWB	37.5	40.9	46.5	52.7	57.2	59.9	61.4	59.6	56.7	52.1	45.2	39.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.8	48.7	54.3	59.0	62.9	67.0	68.3	67.2	63.4	57.5	53.2	45.8
		MCDB	52.6	56.1	66.8	77.0	83.6	89.3	91.9	90.3	85.6	79.6	67.2	56.9
	2%	WB	42.9	46.2	52.0	57.1	60.6	64.7	66.2	64.7	61.1	55.8	50.4	43.7
		MCDB	50.4	53.3	64.6	75.1	82.7	88.1	90.1	88.6	84.2	77.2	64.0	53.3
	5%	WB	40.8	44.0	50.1	55.5	59.3	63.0	64.5	63.0	59.2	54.4	48.5	41.9
		MCDB	47.8	51.8	62.3	72.5	80.4	86.5	88.9	87.3	82.7	75.1	60.3	50.5
	10%	WB	38.4	42.0	48.1	54.0	58.1	61.4	62.9	61.3	57.6	52.8	46.5	40.1
		MCDB	45.2	49.8	59.7	69.2	78.6	84.8	87.7	85.4	81.1	72.1	57.3	47.8
Mean Daily Temperature Range	5% DB	MDBR	18.4	18.9	20.6	21.0	21.3	20.9	19.3	20.4	24.9	24.9	21.3	19.2
		MCDBR	22.1	23.6	25.9	24.7	24.7	23.4	22.1	22.6	27.0	27.2	25.4	24.4
		MCWBR	12.9	12.9	12.2	10.0	9.6	9.7	9.2	10.0	12.5	13.1	13.8	14.2
	5% WB	MCDBR	19.4	19.4	21.8	21.7	22.1	21.2	20.6	20.6	24.7	24.9	21.6	20.8
		MCWBR	12.1	11.6	10.9	9.1	9.4	10.2	10.4	10.3	12.5	11.9	12.1	12.6
Clear-Sky Solar Irradiance	taub		0.306	0.343	0.409	0.450	0.466	0.442	0.434	0.395	0.360	0.375	0.341	0.303
	taud		2.379	2.255	2.047	1.976	1.954	2.027	2.057	2.170	2.252	2.179	2.279	2.408
	Ebn at Noon		283	283	271	266	262	268	269	278	284	266	265	277
	Edh at Noon		29	37	50	57	59	55	53	46	40	40	32	27
All-Sky Solar Radiation	RadAvg		873	1142	1517	1911	2268	2590	2546	2374	1999	1492	1039	832
	RadStd		89	125	99	132	119	75	41	62	49	56	74	79

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.90	N/A	N/A	+1.58	-1.39	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.72	N/A	-3.22	+0.81	-2.12	-2.07	N/A	N/A	N/A	+160

Nomenclature: See separate page